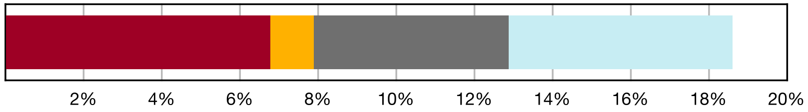
EXTENSION 1
Stage 6

Vectors (Ext1), V1 Introduction to Vectors (Y12)
Vectors and Geometry (Ext1)

Teacher:

Exam Equivalent Time: 78 minutes *(based on allocation of 1.5 minutes per mark)*

V1 Vectors



- Operations with Vectors
- Vectors, Force and Velocity
- Vectors and Geometry
- Vectors and Projectile Motion

***SmarterMaths analytics based on the average contribution to Extension 1 exams since 2020.**

HISTORICAL CONTRIBUTION

- *Vectors* has contributed a highly significant average of 18.6% per new syllabus Ext1 exam since its introduction in 2020.
- This topic has been split into four categories for analysis purposes: *1-Operations with Vectors (6.8%), 2-Vectors, Force and Velocity (1.1%), 3-Vectors and Geometry (5.0%), and 4-Vectors and Projectile Motion (5.7%).*
- This analysis looks at *Vectors and Geometry.*

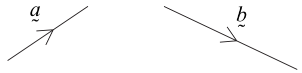
HSC ANALYSIS - What to expect and common pitfalls

- *Vectors and Geometry (5.0%)* has appeared in every new syllabus exam to date, with longer answer questions worth 3–7 marks in 2021–23.
- This topic area can be examined across a very broad difficulty spectrum. While the 2020 exam threw up a low difficulty multiple choice question, the 2023 and 2021 exams included questions that can accurately be described as beasts that defined the best papers.
- This topic requires a deep understanding of the geometry underlying vector addition and subtraction in two planes.
- The question database addresses geometric proofs specifically mentioned in the syllabus, including; proving that rhombus diagonals bisect, mid-points of quadrilaterals form a parallelogram and the relationship between the lengths of the sides and diagonals of a parallelogram.
- Numerous other geometric proofs and results are tested, some of which look at intersections in a given ratio, circular geometry (examined in 2022) and triangular properties.

Questions

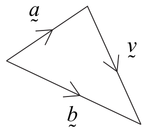
1. Vectors, EXT1 V1 2020 HSC 6 MC

The vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are shown.

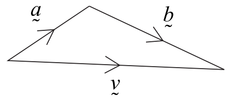


Which diagram below shows the vector $\underset{\sim}{v} = \underset{\sim}{a} - \underset{\sim}{b}$?

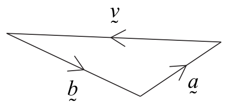
A.



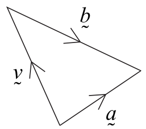
B.



C.

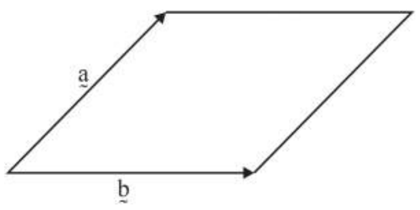


D.



2. Vectors, EXT1 V1 2011 VCE 10 MC

The diagram below shows a rhombus, spanned by the two vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$.



It follows that

A. $\underset{\sim}{a} \cdot \underset{\sim}{b} = 0$

B. $\underset{\sim}{a} = \underset{\sim}{b}$

C. $\left(\underset{\sim}{a} + \underset{\sim}{b}\right) \cdot \left(\underset{\sim}{a} - \underset{\sim}{b}\right) = 0$

D. $\left|\underset{\sim}{a} + \underset{\sim}{b}\right| = \left|\underset{\sim}{a} - \underset{\sim}{b}\right|$

3. Vectors, EXT1 V1 2020 HSC 9 MC

The projection of the vector $\begin{pmatrix} 6 \\ 7 \end{pmatrix}$ onto the line $y = 2x$ is $\begin{pmatrix} 4 \\ 8 \end{pmatrix}$.

The point $(6, 7)$ is reflected in the line $y = 2x$ to a point A.

What is the position vector of the point A?

A. $\begin{pmatrix} 6 \\ 12 \end{pmatrix}$

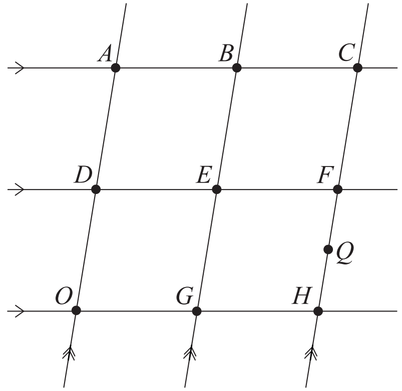
B. $\begin{pmatrix} 2 \\ 9 \end{pmatrix}$

C. $\begin{pmatrix} -6 \\ 7 \end{pmatrix}$

D. $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$

4. Vectors, EXT1 V1 EQ-Bank 4 MC

The diagram shows a grid of equally spaced lines. The vector $\overrightarrow{OA} = \underset{\sim}{a}$ and the vector $\overrightarrow{OH} = \underset{\sim}{h}$. The point Q is halfway between F and H.



Which expression represents the vector $\overrightarrow{EQ}$?

A. $-\frac{1}{4}\underset{\sim}{a} + \frac{1}{2}\underset{\sim}{h}$

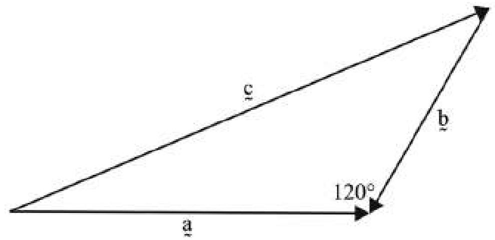
B. $\frac{1}{2}\underset{\sim}{a} - \frac{1}{4}\underset{\sim}{h}$

C. $\frac{1}{4}\underset{\sim}{a} + \frac{1}{2}\underset{\sim}{h}$

D. $\frac{1}{4}\underset{\sim}{a} + \underset{\sim}{h}$

5. Vectors, EXT1 V1 SPEC2 2009 17 MC

Vectors $\underset{\sim}{a}, \underset{\sim}{b}$ and $\underset{\sim}{c}$ are shown below.



From the diagram it follows that

A. $\left|\underset{\sim}{c}\right|^2 = \left|\underset{\sim}{a}\right|^2 + \left|\underset{\sim}{b}\right|^2$

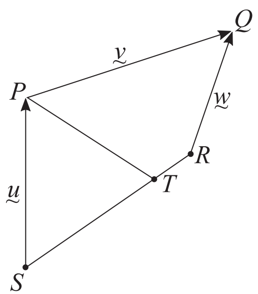
B. $\left|\underset{\sim}{c}\right|^2 = \left|\underset{\sim}{a}\right|^2 + \left|\underset{\sim}{b}\right|^2 - \left|\underset{\sim}{a}\right|\left|\underset{\sim}{b}\right|$

C. $\left|\underset{\sim}{c}\right|^2 = \left|\underset{\sim}{a}\right|^2 + \left|\underset{\sim}{b}\right|^2 + \left|\underset{\sim}{a} \cdot \underset{\sim}{b}\right|$

D. $\left|\underset{\sim}{c}\right|^2 = \left|\underset{\sim}{a}\right|^2 + \left|\underset{\sim}{b}\right|^2 + \left|\underset{\sim}{a}\right|\left|\underset{\sim}{b}\right|$

6. Vectors, EXT1 V1 SM-Bank 10

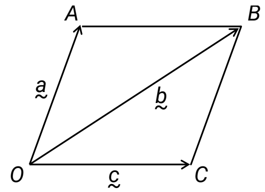
In the quadrilateral $PQRS$, T lies on SR such that $ST:TR = 3:1$.



i. Find $\overrightarrow{TS}$ in terms of $\underset{\sim}{u}$, $\underset{\sim}{v}$ and $\underset{\sim}{w}$. ***(1 mark)***

ii. Hence, find $\overrightarrow{TP}$ in terms of $\underset{\sim}{u}$, $\underset{\sim}{v}$ and $\underset{\sim}{w}$. ***(1 mark)***

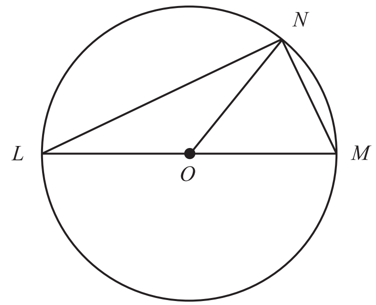
7. Vectors, EXT1 V1 EQ-Bank 27



$OABC$ are the vertices of a rhombus.

Prove, using vectors, that its diagonals are perpendicular. ***(2 marks)***

8. Vectors, EXT1 V1 SM-Bank 2



In the diagram above, LOM is a diameter of the circle with centre O.

N is a point on the circumference of the circle.

If $\underset{\sim}{r} = \overrightarrow{ON}$ and $\underset{\sim}{s} = \overrightarrow{MN}$, express $\overrightarrow{LN}$ in terms of $\underset{\sim}{r}$ and $\underset{\sim}{s}$. ***(2 marks)***

9. Vectors, EXT1 V1 2021 SPEC2 11

Let $\underset{\sim}{i}$ be a unit vector pointing east and let $\underset{\sim}{j}$ be a unit vector pointing north.

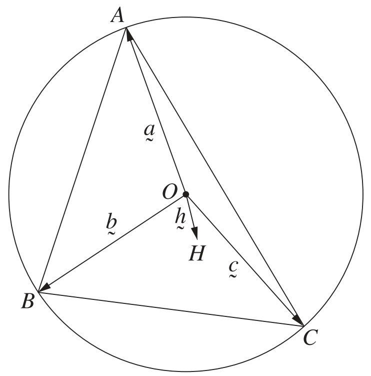
A group of hikers travels 5 km in the direction south 30° west and then north for 10 km before stopping for lunch at position vector $\left(a\underset{\sim}{i}, b\underset{\sim}{j}\right)$.

Find the values of a and b. ***(2 marks)***

10. Vectors, EXT1 V1 2022 HSC 13a

Three different points A, B and C are chosen on a circle centred at O.

Let $\underset{\sim}{a} = \overrightarrow{OA}$, $\underset{\sim}{b} = \overrightarrow{OB}$ and $\underset{\sim}{c} = \overrightarrow{OC}$. Let $\underset{\sim}{h} = \underset{\sim}{a} + \underset{\sim}{b} + \underset{\sim}{c}$ and let H be the point such that $\overrightarrow{OH} = \underset{\sim}{h}$, as shown in the diagram.



NOT
TO
SCALE

Show that $\overrightarrow{BH}$ and $\overrightarrow{CA}$ are perpendicular. ***(3 marks)***

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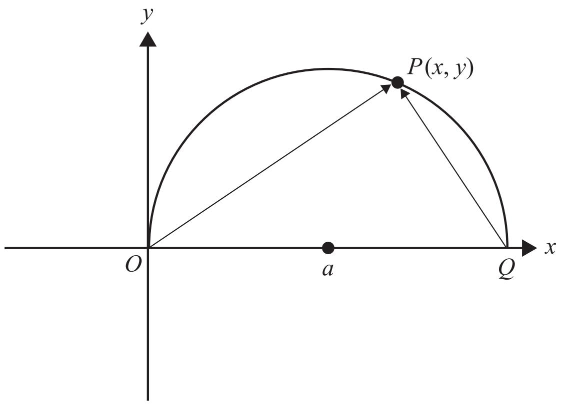
11. Vectors, EXT1 V1 2022 SPEC1 6b

OPQ is a semicircle of radius a with equation $y = \sqrt{a^2 - (x-a)^2}$. $P(x, y)$ is a point on the semicircle OPQ, as shown below.



i. Express the vectors $\overrightarrow{OP}$ and $\overrightarrow{QP}$ in terms of a, x, y, $\underset{\sim}{i}$ and $\underset{\sim}{j}$, where $\underset{\sim}{i}$ is a unit vector in the direction of the positive x-axis and $\underset{\sim}{j}$ is a unit vector in the direction of the positive y-axis. ***(1 mark)***

ii. Hence, using the vector scalar (dot) product, determine whether $\overrightarrow{OP}$ is perpendicular to $\overrightarrow{QP}$. ***(3 marks)***

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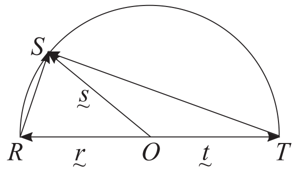
12. Vectors, EXT1 V1 EQ-Bank 2

In the diagram below, ROT is a diameter of the circle with centre O.

S is a point on the circumference.



Using the properties of vectors $\underset{\sim}{r}, \underset{\sim}{s}$ and $\underset{\sim}{t}$, show that $\angle RST$ is a right angle. ***(2 marks)***

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13. Vectors, EXT1 V1 EQ-Bank 6

Point C lies on AB such that $\overrightarrow{AC} = \lambda\overrightarrow{AB}$.

i. Express $\underset{\sim}{c}$ in terms of $\underset{\sim}{a}$ and $\underset{\sim}{b}$. ***(2 marks)***

ii. Hence or otherwise, show that $\overrightarrow{BC} = (1-\lambda)\left(\underset{\sim}{a} - \underset{\sim}{b}\right)$. ***(1 mark)***

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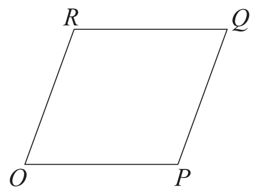
14. Vectors, EXT1 V1 SM-Bank 22

Consider the rhombus $OPQR$ shown below, where $\overrightarrow{OP} = p\underset{\sim}{i}$ and $\overrightarrow{OR} = \underset{\sim}{i} + 2\underset{\sim}{j}$ and p is a real constant.



i. Find p. ***(1 mark)***

ii. Show that the diagonals of rhombus $OPQR$ are perpendicular. ***(2 marks)***

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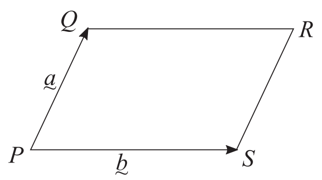
15. Vectors, EXT1 V1 EQ-Bank 24

$PQRS$ is a parallelogram, where $\overrightarrow{PQ} = \underset{\sim}{a}$ and $\overrightarrow{PS} = \underset{\sim}{b}$



Prove, using vectors, that the sum of the squares of the lengths of the diagonals is equal to the sum of the squares of the lengths of the sides. ***(3 marks)***

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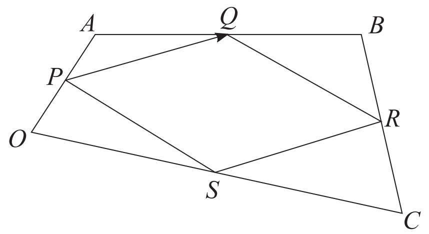
16. Vectors, EXT1 V1 EQ-Bank 26

$OABC$ is a quadrilateral.

P, Q, R and S divide each side of the quadrilateral in half as shown below.



Prove, using vectors, that $PQRS$ is a parallelogram. ***(3 marks)***

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17. Vectors, EXT1 V1 SM-Bank 31

The sum of two unit vectors is a unit vector.

Determine the magnitude of the difference of the two vectors. ***(3 marks)***

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18. Vectors, EXT1 V1 SM-Bank 5

Points A and B are on the number plane. The vector $\overrightarrow{AB}$ is $\begin{pmatrix} 4 \\ 1 \end{pmatrix}$.

Point C is chosen so that the area of ΔABC is $\dfrac{17}{2}$ square units and $\left|\overrightarrow{AC}\right| = \sqrt{34}$.

Find all possible vectors $\overrightarrow{AC}$. ***(4 marks)***

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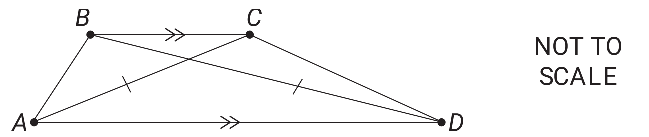
19. Vectors, EXT1 V1 2021 HSC 14c

i. For vector $\underset{\sim}{v}$, show that $\underset{\sim}{v} \cdot \underset{\sim}{v} = \left|\underset{\sim}{v}\right|^2$. ***(1 mark)***

ii. In the trapezium $ABCD$, BC is parallel to AD and $\left|\overrightarrow{AC}\right| = \left|\overrightarrow{BD}\right|$.



Let $\underset{\sim}{a} = \overrightarrow{AB}$, $\underset{\sim}{b} = \overrightarrow{BC}$ and $\overrightarrow{AD} = k\overrightarrow{BC}$, where $k > 0$.

Using part (i), or otherwise, show $2\underset{\sim}{a} \cdot \underset{\sim}{b} + (1-k)\left|\underset{\sim}{b}\right|^2 = 0$. ***(3 marks)***

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20. Vectors, EXT1 V1 2023 HSC 14c

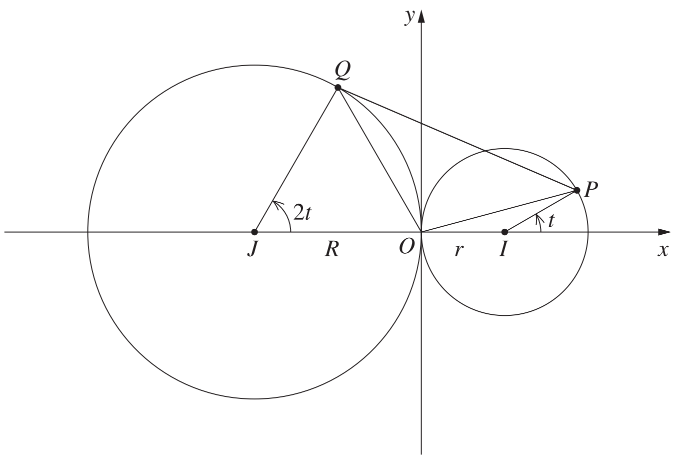
i. Given a non-zero vector $\begin{pmatrix} p \\ q \end{pmatrix}$, it is known that the vector $\begin{pmatrix} q \\ -p \end{pmatrix}$ is perpendicular to $\begin{pmatrix} p \\ q \end{pmatrix}$ and has the same magnitude. (Do NOT prove this.)

Points A and B have position vectors $\overrightarrow{OA} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$, respectively.

Using the given information, or otherwise, show that the area of triangle OAB is $\dfrac{1}{2}|a_1b_2 - a_2b_1|$. ***(3 marks)***

ii. The point P lies on the circle centred at $I(r,0)$ with radius $r > 0$, such that $\overrightarrow{IP}$ makes an angle of t to the horizontal.

The point Q lies on the circle centred at $J(-R,0)$ with radius $R > 0$, such that $\overrightarrow{JQ}$ makes an angle of $2t$ to the horizontal.



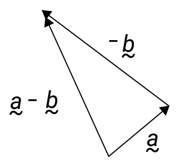
Note that $\overrightarrow{OP} = \overrightarrow{OI} + \overrightarrow{IP}$ and $\overrightarrow{OQ} = \overrightarrow{OJ} + \overrightarrow{JQ}$.

Using part (i), or otherwise, find the values of t, where $-\pi \le t \le \pi$, that maximise the area of triangle OPQ. ***(4 marks)***

Worked Solutions

1. Vectors, EXT1 V1 2020 HSC 6 MC

$\underset{\sim}{v} = \underset{\sim}{a} - \underset{\sim}{b}$



$\Rightarrow D$

2. Vectors, EXT1 V1 2011 VCE 10 MC

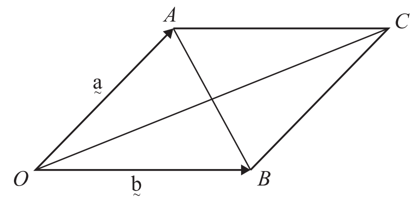
Consider A:

If $\underset{\sim}{a} \cdot \underset{\sim}{b} = 0 \ \Rightarrow \ \underset{\sim}{a} \perp \underset{\sim}{b}$ (incorrect)

Consider B:

$\underset{\sim}{a} \ne \underset{\sim}{b}$ (incorrect)

Consider C:



$\underset{\sim}{a} + \underset{\sim}{b} = \overrightarrow{OC}$

$\underset{\sim}{a} - \underset{\sim}{b} = \overrightarrow{BA}$

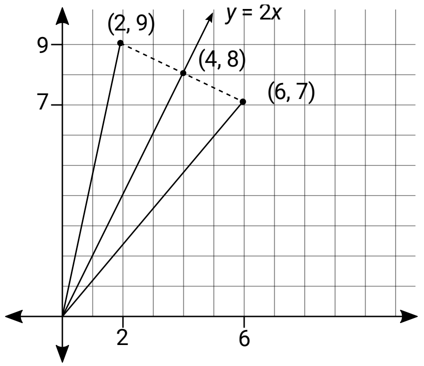
$\overrightarrow{OC} \cdot \overrightarrow{BA} = 0$

The diagonals of a rhombus are perpendicular (correct)

$\Rightarrow C$

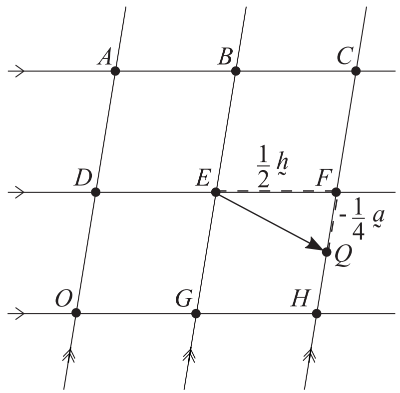
3. Vectors, EXT1 V1 2020 HSC 9 MC

Graph the projection and reflection:



$\Rightarrow B$

4. Vectors, EXT1 V1 EQ-Bank 4 MC



$\overrightarrow{EQ} = \frac{1}{2}\underset{\sim}{h} - \frac{1}{4}\underset{\sim}{a}$

$\Rightarrow A$

5. Vectors, EXT1 V1 SPEC2 2009 17 MC

$\underset{\sim}{c} + \underset{\sim}{b} = \underset{\sim}{a} \quad \Rightarrow \quad \underset{\sim}{c} = \underset{\sim}{a} - \underset{\sim}{b}$

$\therefore \underset{\sim}{c} \cdot \underset{\sim}{c} = \left(\underset{\sim}{a} - \underset{\sim}{b}\right) \cdot \left(\underset{\sim}{a} - \underset{\sim}{b}\right)$

$= \underset{\sim}{a} \cdot \underset{\sim}{a} - \underset{\sim}{a} \cdot \underset{\sim}{b} - \underset{\sim}{b} \cdot \underset{\sim}{a} + \underset{\sim}{b} \cdot \underset{\sim}{b}$

$= \underset{\sim}{a} \cdot \underset{\sim}{a} + \underset{\sim}{b} \cdot \underset{\sim}{b} - 2\underset{\sim}{b} \cdot \underset{\sim}{a}$

$\therefore \left|\underset{\sim}{c}\right|^2 = \left|\underset{\sim}{a}\right|^2 + \left|\underset{\sim}{b}\right|^2 - 2\left|\underset{\sim}{a}\right|\left|\underset{\sim}{b}\right| \cos 120^\circ$

$\therefore \left|\underset{\sim}{c}\right|^2 = \left|\underset{\sim}{a}\right|^2 + \left|\underset{\sim}{b}\right|^2 + \left|\underset{\sim}{a}\right|\left|\underset{\sim}{b}\right|$

$\Rightarrow D$

6. Vectors, EXT1 V1 SM-Bank 10

i. $\overrightarrow{TS} = \frac{3}{4} \times -\overrightarrow{SR}$

$\overrightarrow{SR} = \underset{\sim}{u} + \underset{\sim}{v} - \underset{\sim}{w}$

$\therefore \overrightarrow{TS} = \frac{3}{4} \times -\left(\underset{\sim}{u} + \underset{\sim}{v} - \underset{\sim}{w}\right)$

$= \frac{3}{4}\left(\underset{\sim}{w} - \underset{\sim}{u} - \underset{\sim}{v}\right)$

ii. $\overrightarrow{TP} = \overrightarrow{SP} + \overrightarrow{TS}$

$= \underset{\sim}{u} + \frac{3}{4}\left(\underset{\sim}{w} - \underset{\sim}{u} - \underset{\sim}{v}\right)$

$= \frac{1}{4}\underset{\sim}{u} + \frac{3}{4}\underset{\sim}{w} - \frac{3}{4}\underset{\sim}{v}$

7. Vectors, EXT1 V1 EQ-Bank 27

Since $OABC$ is a rhombus

$\left|\underset{\sim}{a}\right| = \left|\underset{\sim}{c}\right|$

Diagonals are $\overrightarrow{OB}$ and $\overrightarrow{AC}$, where

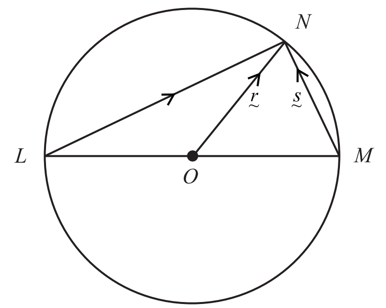
$\overrightarrow{OB} = \underset{\sim}{a} + \underset{\sim}{c}$

$\overrightarrow{AC} = \underset{\sim}{c} - \underset{\sim}{a}$

$$\begin{aligned}\left(\underset{\sim}{c}-\underset{\sim}{a}\right) \cdot \left(\underset{\sim}{a}+\underset{\sim}{c}\right) &= \underset{\sim}{c}\cdot\underset{\sim}{a} + \underset{\sim}{c}\cdot\underset{\sim}{c} - \underset{\sim}{a}\cdot\underset{\sim}{a} - \underset{\sim}{a}\cdot\underset{\sim}{c} \\ &= \left|\underset{\sim}{c}\right|^2 - \left|\underset{\sim}{a}\right|^2 \\ &= 0\end{aligned}$$

$\therefore\ \overrightarrow{OB} \perp \overrightarrow{AC}$

8. Vectors, EXT1 V1 SM-Bank 2



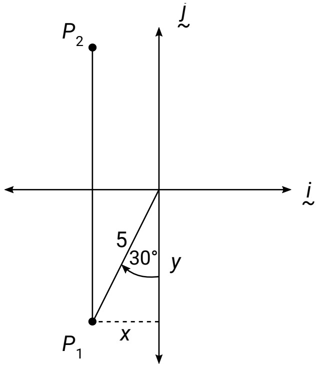
$$\overrightarrow{LN} = \overrightarrow{LM} + \overrightarrow{MN}$$

$$\begin{aligned}\overrightarrow{LN} &= \overrightarrow{LO} + \overrightarrow{ON} \\ &= \frac{1}{2}\overrightarrow{LM} + \overrightarrow{ON}\end{aligned}$$

$$\begin{aligned}\therefore\ \overrightarrow{LM} + \underset{\sim}{s} &= \frac{1}{2}\overrightarrow{LM} + \underset{\sim}{r} \\ \frac{1}{2}\overrightarrow{LM} &= \underset{\sim}{r} - \underset{\sim}{s} \\ \overrightarrow{LM} &= 2\underset{\sim}{r} - 2\underset{\sim}{s}\end{aligned}$$

$$\begin{aligned}\therefore\ \overrightarrow{LN} &= 2\underset{\sim}{r} - 2\underset{\sim}{s} + \underset{\sim}{s} \\ &= 2\underset{\sim}{r} - \underset{\sim}{s}\end{aligned}$$

9. Vectors, EXT1 V1 2021 SPEC2 11



$\sin 30 = \dfrac{x}{5} \Rightarrow x = \dfrac{5}{2}$

$\cos 30 = \dfrac{y}{5} \Rightarrow y = \dfrac{5\sqrt{3}}{2}$

P_1 vector: $\left(-\dfrac{5}{2}\underset{\sim}{i}, -\dfrac{5\sqrt{3}}{2}\underset{\sim}{j}\right)$

P_2 vector: $\left(-\dfrac{5}{2}\underset{\sim}{i}, \left(10-\dfrac{5\sqrt{3}}{2}\right)\underset{\sim}{j}\right)$

$\therefore a = -\dfrac{5}{2}, b = 10-\dfrac{5\sqrt{3}}{2}$

10. Vectors, EXT1 V1 2022 HSC 13a

Prove $\overrightarrow{BH} \perp \overrightarrow{CA}$:

$\overrightarrow{BH} \cdot \overrightarrow{CA} = \left(-\underset{\sim}{b} + \underset{\sim}{h}\right) \cdot \left(-\underset{\sim}{c} + \underset{\sim}{a}\right)$

Given $\underset{\sim}{h} = \underset{\sim}{a} + \underset{\sim}{b} + \underset{\sim}{c} \Rightarrow \underset{\sim}{h} - \underset{\sim}{b} = \underset{\sim}{a} + \underset{\sim}{c}$

$$\begin{aligned}\overrightarrow{BH} \cdot \overrightarrow{CA} &= \left(\underset{\sim}{a} + \underset{\sim}{c}\right) \cdot \left(\underset{\sim}{a} - \underset{\sim}{c}\right) \\ &= \left|\underset{\sim}{a}\right|^2 - \left|\underset{\sim}{c}\right|^2\end{aligned}$$

$\underset{\sim}{a}$ and $\underset{\sim}{c}$ are radii $\Rightarrow \left|\underset{\sim}{a}\right| = \left|\underset{\sim}{c}\right|$

$\therefore$ Since $\overrightarrow{BH} \cdot \overrightarrow{CA} = 0 \Rightarrow \overrightarrow{BH} \perp \overrightarrow{CA}$

11. Vectors, EXT1 V1 2022 SPEC1 6b

i. $\overrightarrow{OP} = x\underset{\sim}{i} + \sqrt{a^2 - (x-a)^2}\underset{\sim}{j}$

$\overrightarrow{QP} = (x - 2a)\underset{\sim}{i} + \sqrt{a^2 - (x-a)^2}\underset{\sim}{j}$

ii.
$$\begin{aligned}\overrightarrow{OP} \cdot \overrightarrow{QP} &= x(x-2a) + a^2 - (x-a)^2 \\ &= x^2 - 2ax + a^2 - x^2 + 2ax - a^2 \\ &= 0\end{aligned}$$

$\therefore \overrightarrow{OP}$ and $\overrightarrow{QP}$ are perpendicular.

12. Vectors, EXT1 V1 EQ-Bank 2

$\left|\underset{\sim}{r}\right| = \left|\underset{\sim}{s}\right| = \left|\underset{\sim}{t}\right|$ (radii)

$\underset{\sim}{r} = -\underset{\sim}{t}$

$\overrightarrow{RS} = \underset{\sim}{s} - \underset{\sim}{r}$

$\overrightarrow{TS} = \underset{\sim}{s} - \underset{\sim}{t}$

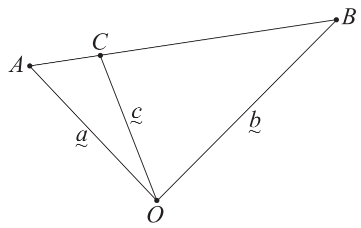
$$\begin{aligned}\overrightarrow{RS} \cdot \overrightarrow{TS} &= \left(\underset{\sim}{s} - \underset{\sim}{r}\right) \cdot \left(\underset{\sim}{s} - \underset{\sim}{t}\right) \\ &= \left(\underset{\sim}{s} - \underset{\sim}{r}\right) \cdot \left(\underset{\sim}{s} + \underset{\sim}{r}\right) \quad \left(\text{using } \underset{\sim}{r} = -\underset{\sim}{t}\right) \\ &= \underset{\sim}{s} \cdot \left(\underset{\sim}{s} + \underset{\sim}{r}\right) - \underset{\sim}{r} \cdot \left(\underset{\sim}{s} + \underset{\sim}{r}\right) \\ &= \underset{\sim}{s} \cdot \underset{\sim}{s} + \underset{\sim}{s} \cdot \underset{\sim}{r} - \underset{\sim}{r} \cdot \underset{\sim}{s} - \underset{\sim}{r} \cdot \underset{\sim}{r} \\ &= \left|\underset{\sim}{s}\right|^2 - \left|\underset{\sim}{r}\right|^2 \\ &= 0\end{aligned}$$

$\therefore \overrightarrow{RS} \perp \overrightarrow{TS}$

$\therefore \angle RST$ is a right angle.

13. Vectors, EXT1 V1 EQ-Bank 6

i.



$$\underset{\sim}{c} = \underset{\sim}{a} + \overrightarrow{AC}$$

$$\overrightarrow{AB} = \underset{\sim}{b} - \underset{\sim}{a}$$

$$\begin{aligned}\overrightarrow{AC} &= \lambda\overrightarrow{AB}\\ &= \lambda\left(\underset{\sim}{b} - \underset{\sim}{a}\right)\end{aligned}$$

$$\begin{aligned}\therefore \underset{\sim}{c} &= \underset{\sim}{a} + \lambda\left(\underset{\sim}{b} - \underset{\sim}{a}\right)\\ &= \lambda\underset{\sim}{b} + (1-\lambda)\underset{\sim}{a}\end{aligned}$$

ii.

$$\begin{aligned}\overrightarrow{BC} &= \underset{\sim}{c} - \underset{\sim}{b}\\ &= \lambda\underset{\sim}{b} + (1-\lambda)\underset{\sim}{a} - \underset{\sim}{b}\\ &= (1-\lambda)\underset{\sim}{a} + (\lambda-1)\underset{\sim}{b}\\ &= (1-\lambda)\underset{\sim}{a} - (1-\lambda)\underset{\sim}{b}\\ &= (1-\lambda)\left(\underset{\sim}{a} - \underset{\sim}{b}\right)\end{aligned}$$

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14. Vectors, EXT1 V1 SM-Bank 22

i. $\left|\overrightarrow{OP}\right| = \left|\overrightarrow{OR}\right|$

$$\begin{aligned}p &= \sqrt{1^2 + 2^2}\\ &= \sqrt{5}\end{aligned}$$

ii. $\overrightarrow{RQ} = \overrightarrow{OP}, \quad \overrightarrow{OR} = \overrightarrow{PQ}$

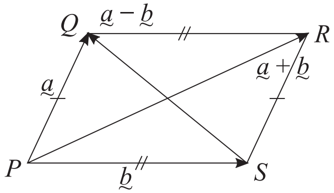
$$\begin{aligned}\overrightarrow{OQ} &= \overrightarrow{OR} + \overrightarrow{RQ}\\ &= \begin{bmatrix}1\\2\end{bmatrix} + \begin{bmatrix}\sqrt{5}\\0\end{bmatrix}\\ &= \left(1+\sqrt{5}\right)\underset{\sim}{i} + 2\underset{\sim}{j}\end{aligned}$$

$$\begin{aligned}\overrightarrow{PR} &= \overrightarrow{OR} - \overrightarrow{RQ}\\ &= \begin{bmatrix}1\\2\end{bmatrix} - \begin{bmatrix}\sqrt{5}\\0\end{bmatrix}\\ &= \left(1-\sqrt{5}\right)\underset{\sim}{i} + 2\underset{\sim}{j}\end{aligned}$$

$$\begin{aligned}\overrightarrow{OQ}\cdot\overrightarrow{PR} &= \begin{bmatrix}1+\sqrt{5}\\2\end{bmatrix}\cdot\begin{bmatrix}1-\sqrt{5}\\2\end{bmatrix}\\ &= \left(1+\sqrt{5}\right)\left(1-\sqrt{5}\right) + 2\times 2\\ &= 1-5+4\\ &= 0\end{aligned}$$

$$\therefore \overrightarrow{OQ} \perp \overrightarrow{PQ}$$

15. Vectors, EXT1 V1 EQ-Bank 24



$$\overrightarrow{PR} = \underset{\sim}{a} + \underset{\sim}{b}, \quad \overrightarrow{SQ} = \underset{\sim}{a} - \underset{\sim}{b}$$

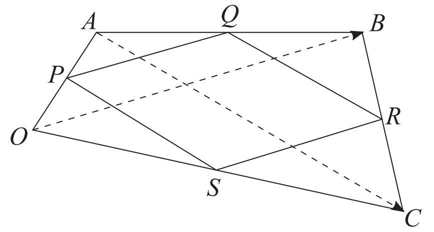
$$\overrightarrow{PQ} = \overrightarrow{SR} = \underset{\sim}{a}$$

$$\overrightarrow{PS} = \overrightarrow{QR} = \underset{\sim}{b}$$

Prove $\left|\underset{\sim}{a} + \underset{\sim}{b}\right|^2 + \left|\underset{\sim}{a} - \underset{\sim}{b}\right|^2 = 2\left|\underset{\sim}{a}\right|^2 + 2\left|\underset{\sim}{b}\right|^2$

$$\begin{aligned}
\left|\underset{\sim}{a} + \underset{\sim}{b}\right|^2 + \left|\underset{\sim}{a} - \underset{\sim}{b}\right|^2 &= \left(\underset{\sim}{a} + \underset{\sim}{b}\right) \cdot \left(\underset{\sim}{a} + \underset{\sim}{b}\right) + \left(\underset{\sim}{a} - \underset{\sim}{b}\right)\left(\underset{\sim}{a} - \underset{\sim}{b}\right) \\
&= \underset{\sim}{a} \cdot \underset{\sim}{a} + \underset{\sim}{b} \cdot \underset{\sim}{b} + 2\underset{\sim}{a} \cdot \underset{\sim}{b} + \underset{\sim}{a} \cdot \underset{\sim}{a} + \underset{\sim}{b} \cdot \underset{\sim}{b} - 2\underset{\sim}{a} \cdot \underset{\sim}{b} \\
&= \left|\underset{\sim}{a}\right|^2 + \left|\underset{\sim}{b}\right|^2 + \left|\underset{\sim}{a}\right|^2 + \left|\underset{\sim}{b}\right|^2 \\
&= 2\left|\underset{\sim}{a}\right|^2 + 2\left|\underset{\sim}{b}\right|^2 \quad \ldots \text{ as required}
\end{aligned}$$

16. Vectors, EXT1 V1 EQ-Bank 26



Consider diagonal $\overrightarrow{OB}$:

$$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OC} + \overrightarrow{CB}$$

$$\overrightarrow{PQ} = \overrightarrow{PA} + \overrightarrow{AQ} = \frac{1}{2}\left(\overrightarrow{OA} + \overrightarrow{AB}\right) = \frac{1}{2}\overrightarrow{OB}$$

$$\overrightarrow{SR} = \overrightarrow{SC} + \overrightarrow{CR} = \frac{1}{2}\left(\overrightarrow{OC} + \overrightarrow{CB}\right) = \frac{1}{2}\overrightarrow{OB}$$

$$\therefore \overrightarrow{PQ} = \overrightarrow{SR}$$

Consider diagonal $\overrightarrow{AC}$:

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AO} + \overrightarrow{OC}$$

$$\overrightarrow{QR} = \overrightarrow{QB} + \overrightarrow{BR} = \frac{1}{2}\left(\overrightarrow{AB} + \overrightarrow{BC}\right) = \frac{1}{2}\overrightarrow{AC}$$

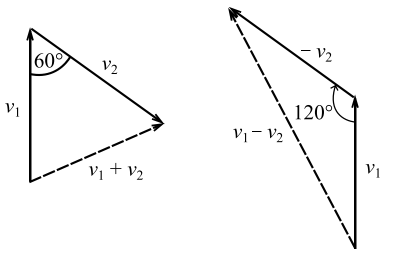
$$\overrightarrow{PS} = \overrightarrow{PO} + \overrightarrow{OS} = \frac{1}{2}\left(\overrightarrow{AO} + \overrightarrow{OC}\right) = \frac{1}{2}\overrightarrow{AC}$$

$$\therefore \overrightarrow{QR} = \overrightarrow{PS}$$

Since $PQRS$ has equal opposite sides,
$PQRS$ is a parallelogram.

17. Vectors, EXT1 V1 SM-Bank 31

Vectors can be drawn as two sides of an equilateral triangle.



Using the cosine rule for the difference between the two vectors:

$c^2 = a^2 + b^2 - 2ab\cos C$

$\quad = 1 + 1 - 2 \times -\dfrac{1}{2}$

$\quad = 3$

$c = \sqrt{3}$

$|v_1 - v_2| = \sqrt{3}$

18. Vectors, EXT1 V1 SM-Bank 5

$$\left|\overrightarrow{AB}\right| = \sqrt{4^2+1^2} = \sqrt{17}$$

Let $\angle CAB = \theta$

$$\text{Area } \Delta ABC = \frac{1}{2}\cdot\left|\overrightarrow{AB}\right|\cdot\left|\overrightarrow{AC}\right|\cdot\sin\theta$$

$$\frac{17}{2} = \frac{1}{2}\cdot\sqrt{17}\cdot\sqrt{34}\sin\theta$$

$$17 = 17\sqrt{2}\sin\theta$$

$$\sin\theta = \frac{1}{\sqrt{2}}$$

$$\theta = \frac{\pi}{4} \text{ or } \frac{3\pi}{4} \quad (\theta > 0)$$

$$\Rightarrow \cos\theta = \pm\frac{1}{\sqrt{2}}$$

$$\overrightarrow{AB}\cdot\overrightarrow{AC} = \left|\overrightarrow{AB}\right|\cdot\left|\overrightarrow{AC}\right|\cdot\cos\theta$$

$$= \pm\sqrt{17}\cdot\sqrt{34}\cdot\frac{1}{\sqrt{2}}$$

$$= \pm 17$$

Let $\overrightarrow{AC} = \begin{pmatrix} x \\ y \end{pmatrix}$

$$\overrightarrow{AB}\cdot\overrightarrow{AC} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

$$= 4x + y$$

$$4x + y = \pm 17 \ldots \ (1)$$

$$\left|\overrightarrow{AC}\right| = \sqrt{x^2+y^2} = \sqrt{34}$$

$$x^2 + y^2 = 34 \ \ldots \ (2)$$

Solving simultaneously:

$4x + y = 17 \ \Rightarrow \ y = 17 – 4x$

Substitute into (2)

$$x^2 + (17–4x)^2 = 34$$

$$x^2 + 289–136x + 16x^2 = 34$$

$$17x^2–136x + 255 = 0$$

$$x^2–8x + 15 = 0$$

$$(x–3)(x–5) = 0$$

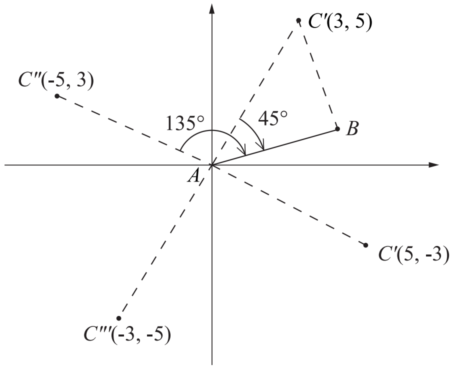
$\therefore x = 3, y = 5 \quad \text{or} \quad x = 5, y = -3$

Similarly for $4x + y = -17 \Rightarrow y = -4x - 17$

$$x^2 + (-4x-17)^2 = 34$$
$$x^2 + 8x + 15 = 0$$
$$(x+3)(x+5) = 0$$

$\therefore x = -3, y = -5 \quad \text{or} \quad x = -5, y = 3$

COMMENT: A diagram is highly recommended. Here, the possibility of 4 solutions is clearly shown.



$\therefore$ Possible vectors $\overrightarrow{AC}$ are

$\begin{pmatrix} 3 \\ 5 \end{pmatrix}, \begin{pmatrix} 5 \\ -3 \end{pmatrix}, \begin{pmatrix} -3 \\ -5 \end{pmatrix}$ and $\begin{pmatrix} -5 \\ 3 \end{pmatrix}$

19. Vectors, EXT1 V1 2021 HSC 14c

i. Let $\underset{\sim}{v} = \begin{pmatrix} x \\ y \end{pmatrix}$

$$\left|\underset{\sim}{v}\right| = \sqrt{x^2 + y^2}$$
$$\underset{\sim}{v} \cdot \underset{\sim}{v} = x^2 + y^2 = \left(\sqrt{x^2 + y^2}\right)^2 = \left|\underset{\sim}{v}\right|^2$$

ii. Show $2\underset{\sim}{a} \cdot \underset{\sim}{b} + (1-k)\left|\underset{\sim}{b}\right|^2 = 0$

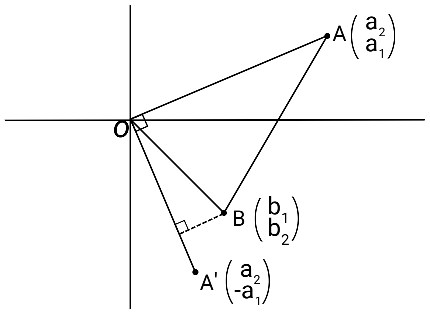
♦♦♦ *Mean mark part (ii) 18%.*

$$\left|\overrightarrow{AC}\right| = \left|\overrightarrow{BD}\right|$$
$$\left|\underset{\sim}{a} + \underset{\sim}{b}\right| = \left|k\underset{\sim}{b} - \underset{\sim}{a}\right|$$
$$\left|\underset{\sim}{a} + \underset{\sim}{b}\right|^2 = \left|k\underset{\sim}{b} - \underset{\sim}{a}\right|^2$$
$$\left(\underset{\sim}{a} + \underset{\sim}{b}\right)\left(\underset{\sim}{a} + \underset{\sim}{b}\right) = \left(k\underset{\sim}{b} - \underset{\sim}{a}\right)\left(k\underset{\sim}{b} - \underset{\sim}{a}\right) \quad \text{(see part a.)}$$
$$\underset{\sim}{a} \cdot \underset{\sim}{a} + 2\underset{\sim}{a} \cdot \underset{\sim}{b} + \underset{\sim}{b} \cdot \underset{\sim}{b} = k^2\underset{\sim}{b} \cdot \underset{\sim}{b} - 2k\underset{\sim}{a} \cdot \underset{\sim}{b} + \underset{\sim}{a} \cdot \underset{\sim}{a}$$
$$2\underset{\sim}{a} \cdot \underset{\sim}{b} + 2k\underset{\sim}{a} \cdot \underset{\sim}{b} + \underset{\sim}{b} \cdot \underset{\sim}{b} - k^2\underset{\sim}{b} \cdot \underset{\sim}{b} = 0$$
$$2\underset{\sim}{a} \cdot \underset{\sim}{b}(1+k) + \underset{\sim}{b} \cdot \underset{\sim}{b}(1-k^2) = 0$$
$$2\underset{\sim}{a} \cdot \underset{\sim}{b}(1+k) + \underset{\sim}{b} \cdot \underset{\sim}{b}(1+k)(1-k) = 0$$
$$2\underset{\sim}{a} \cdot \underset{\sim}{b} + (1-k)\left|\underset{\sim}{b}\right|^2 = 0$$

20. Vectors, EXT1 V1 2023 HSC 14c

i.

♦♦♦ Mean mark (i) 18%.



$$\left|\text{proj}_{\overrightarrow{OA'}}\overrightarrow{OB}\right| = \perp \text{ height of } \triangle OAB \text{ from side } \overrightarrow{OA}$$

$$= \left| \frac{\overrightarrow{OB} \cdot \overrightarrow{OA'}}{|\overrightarrow{OA'}|} \right|$$

$$= \left| \frac{b_1a_2 - b_2a_1}{|\overrightarrow{OA'}|} \right|$$

$$\text{Area } \triangle AOB = \frac{1}{2} \left| \frac{b_1a_2 - b_2a_1}{|\overrightarrow{OA'}|} \right| \cdot |\overrightarrow{OA}|$$

$$= \frac{1}{2}|a_1b_2 - a_2b_1| \quad (\text{noting } \ |\overrightarrow{OA}| = |\overrightarrow{OA'}|)$$

♦♦♦ Mean mark (ii) 8%.

ii. $\overrightarrow{OP} = \overrightarrow{OI} + \overrightarrow{IP}$

$$= \begin{pmatrix} r \\ 0 \end{pmatrix} + \begin{pmatrix} r\cos t \\ r\sin t \end{pmatrix}$$

$$= \begin{pmatrix} r(1+\cos t) \\ r\sin t \end{pmatrix}$$

$$\overrightarrow{OQ} = \overrightarrow{OJ} + \overrightarrow{JQ}$$

$$= \begin{pmatrix} -R \\ 0 \end{pmatrix} + \begin{pmatrix} R\cos 2t \\ R\sin 2t \end{pmatrix}$$

$$= \begin{pmatrix} R(\cos 2t - 1) \\ R\sin 2t \end{pmatrix}$$

Using part (i):

$A_{\triangle OPQ}$

$$= \frac{1}{2}\left| r(1+\cos t)\cdot R\sin 2t - r\sin t \cdot R(\cos 2t - 1) \right|$$

$$= \frac{rR}{2}\left| \sin 2t + \cos t \sin 2t - \sin t \cos 2t + \sin t \right|$$

$$= \frac{rR}{2}\left| \sin 2t + \sin t + \sin(2t - t) \right|$$

$$= \frac{rR}{2}\left| 2\sin t\cos t + 2\sin t \right|$$

$$= rR\left| \sin t(\cos t + 1) \right|$$

Let $f(t) = \sin t(\cos t + 1)$

$$f'(t) = \cos t(\cos t + 1) + \sin t(-\sin t)$$
$$= \cos^2 t + \cos t - \sin^2 t$$
$$= \cos^2 t + \cos t - (1 - \cos^2 t)$$
$$= 2\cos^2 t + \cos t - 1$$
$$= (2\cos^2 t - 1)(\cos t + 1)$$

SP's when $f'(t) = 0$:

$\cos t = \dfrac{1}{2}$ $\qquad$ $\cos t = -1$

$t = \dfrac{\pi}{3}, -\dfrac{\pi}{3}$ $\qquad$ $t = \pi, \ -\pi$

Testing SP's:

$f(\pi) = f(-\pi) = 0$

$\therefore A_{\triangle AOB}$ is maximum when $t = \dfrac{\pi}{3}, -\dfrac{\pi}{3}$